

The relative position of a line and a plane

Three cases, one test — and the theorem that turns a hard space question into an easy plane one.

LESSON 23–24

2 × 40 MIN

GEOMETRY 10, §11

EXTENSION

LESSON CLOCK

8'	22'	24'	22'	18'	6'
The three cases			8 min		
The test			22 min		
The converse theorem			24 min		
Applications in a cube			22 min		
Practice			18 min		
Homework			6 min		

LEARNING OBJECTIVES

- List the three positions of a line and a plane.
- State and apply the test for a line parallel to a plane.
- Prove that a plane through a line parallel to another meets it in a parallel line.
- Use the results in a cube or prism.

TEXTBOOK

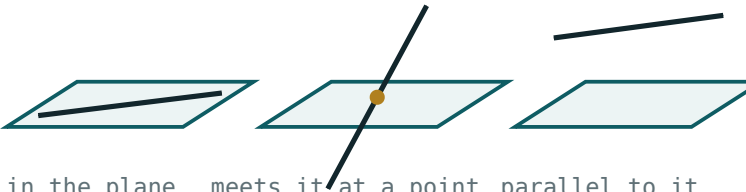
National	pp. 103–112
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 11.1

01

Explanation

The three cases

a line and a plane



in the plane meets it at a point parallel to it

In the plane, meeting it at a point, parallel to it. There is no fourth case.

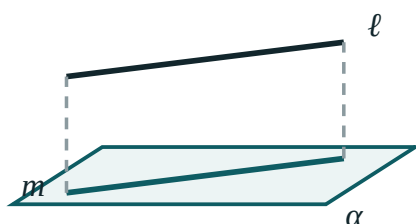
CASE	COMMON POINTS	NOTATION
the line lies in the plane	infinitely many	$a \subset \alpha$
the line meets the plane	1	$a \cap \alpha = M$
the line is parallel to the plane	0	$a \parallel \alpha$

Two common points would put the whole line in the plane, by axiom S2 — which is why there is no case with exactly two.

The test

TEST FOR A LINE PARALLEL TO A PLANE

If a line ℓ not lying in a plane α is parallel to **some** line m lying in α , then $\ell \parallel \alpha$.



$\ell \parallel m$ and m lies in α , so $\ell \parallel \alpha$

ℓ is parallel to m , and m lies in α . That is enough.

Proof. ℓ and m are parallel, so they lie in one plane β . If ℓ met α , the meeting point would lie in $\beta \cap \alpha = m$ — so it would be on m , contradicting $\ell \parallel m$. ■

“SOME”, NOT “EVERY”

You need to exhibit **one** line of the plane parallel to ℓ . Finding it is the whole exercise; in a cube it is almost always the opposite edge of a face.

The converse

Theorem. If $\ell \parallel \alpha$ and a plane β contains ℓ and meets α in a line m , then $\ell \parallel m$.

Proof. ℓ and m both lie in β , so they are coplanar. If they met, the meeting point would be on ℓ and in α — impossible, since $\ell \parallel \alpha$. ■

THIS IS THE SECTION RULE R2, PROVED

It is why a cutting plane meets two parallel faces in parallel lines: each face is parallel to the other, and the cutting plane contains a line of each.

In a cube

In $ABCD A_1 B_1 C_1 D_1$, is $A_1 B_1$ parallel to the plane $ABCD$?

1. $A_1 B_1$ does not lie in $ABCD$.
2. $A_1 B_1 \parallel AB$, opposite sides of the square face $ABB_1 A_1$.
3. $AB \subset ABCD$.
4. By the test, $A_1 B_1 \parallel ABCD$. ■

The **distance** from a line parallel to a plane to that plane is the same at every point — here it is the height of the cube. That constancy is what makes the distance well defined.

02 Worked examples

EXAMPLE 1 Prove that $B_1 C_1$ is parallel to the plane $ABCD$ in a cube.

① $B_1 C_1 \not\subset ABCD$

It is on the top face.

② $B_1 C_1 \parallel BC$

Opposite sides of face $BCC_1 B_1$.

③ $BC \subset ABCD$

④ Test applies.

ANSWER

$B_1 C_1 \parallel ABCD$

EXAMPLE 2 A plane through A_1B_1 meets the base $ABCD$ in a line m . What can be said about m ?

① $A_1B_1 \parallel ABCD$

By the previous example.

② The cutting plane contains A_1B_1 and meets the base in m .

③ By the converse theorem, $m \parallel A_1B_1$.

ANSWER

$m \parallel A_1B_1$, so also $m \parallel AB$

EXAMPLE 3 Is the diagonal AC_1 of a cube parallel to any face?

① It meets the base at A and the top at C_1 .

② So it is not parallel to those two.

③ It meets each of the other four faces too.
It is a space diagonal.

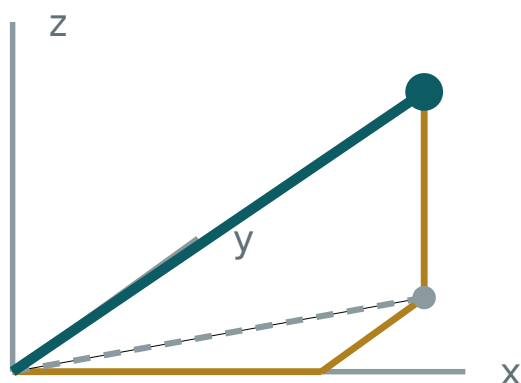
ANSWER

No — a space diagonal meets every face

03 Interactive model

Hold a pencil parallel to the table and ask which line on the table it is parallel to.

Line and plane



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Line parallel to a plane	Tekislikka parallel chiziq	Прямая, параллельная плоскости
Test for parallelism	Parallellik alomati	Признак параллельности
Line lying in a plane	Tekislikda yotuvchi chiziq	Прямая, лежащая в плоскости
Point of intersection	Kesishish nuqtasi	Точка пересечения
Auxiliary plane	Yordamchi tekislik	Вспомогательная плоскость
Trace	Iz	След
Distance from a line to a plane	Chiziqdan tekislikkacha masofa	Расстояние от прямой до плоскости
Section	Kesim	Сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A line and a plane can share:

0, 1 or infinitely many points

exactly 2

only 0

only 1

The test for $\ell \parallel \alpha$ needs:

every line of α

one line of α parallel to ℓ

a perpendicular

a point

If $\ell \parallel \alpha$ and $\beta \supset \ell$ cuts α in m , then:

$\ell \perp m$

$\ell \parallel m$

$\ell = m$

nothing

A_1B_1 and the plane $ABCD$ in a cube are:

meeting

parallel

coincident

perpendicular

A space diagonal of a cube is parallel to:

the base

a side face

no face

every face

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the three positions of a line and a plane

ANSWER in it, meeting it, parallel

2. Can a line and a plane share exactly 2 points?

ANSWER no

3. State the test for $\ell \parallel \alpha$

ANSWER $\ell \parallel m$ for some $m \subset \alpha$

4. Is $A_1B_1 \parallel ABCD$ in a cube?

ANSWER yes

5. Is $AA_1 \parallel ABCD$?

ANSWER no — it meets it at A

6. Is AC_1 parallel to any face?

ANSWER no

7. Distance from a parallel line to a plane is:

ANSWER constant

Medium · the standard question

7 PROBLEMS

1. Prove $B_1C_1 \parallel ABCD$

ANSWER It is parallel to BC , which lies in the base

2. Prove $A_1C_1 \parallel ABCD$

ANSWER $A_1C_1 \parallel AC \subset ABCD$

3. Which faces is AB parallel to?

ANSWER $A_1B_1C_1D_1$ and DCC_1D_1

4. Which faces does AB lie in?

ANSWER $ABCD$ and ABB_1A_1

5. Which faces does AB meet at a point?

ANSWER ADD_1A_1 and BCC_1B_1

6. A plane through AB meets the top face in m . Then m

ANSWER $\parallel AB$

7. Distance from A_1B_1 to the base in a cube of side 7

ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. Prove the test for a line parallel to a plane

ANSWER A meeting point would lie on the line of intersection

2. Prove the converse theorem

ANSWER ℓ and m are coplanar in β and cannot meet

3. Through a point off a plane, how many lines are parallel to the plane?

ANSWER infinitely many — a whole plane of them

4. Prove: if $\ell \parallel \alpha$ and $\ell \parallel \beta$ with $\alpha \cap \beta = m$, then $\ell \parallel m$

ANSWER Apply the converse in each plane

5. In a cube, find a plane parallel to AC but not containing it

ANSWER $A_1B_1C_1D_1$

6. Give a line parallel to two intersecting planes

ANSWER One parallel to their line of intersection

7. Why can a line parallel to a plane never meet a line of that plane?

ANSWER The meeting point would be common to line and plane

07 Homework

Homework — 5 tasks

Each proof must name the line of the plane it uses.

1. Prove that D_1C_1 is parallel to the plane $ABCD$.
2. List every face of a cube that AD is parallel to.
3. A plane through B_1C_1 meets the base in a line m . What is m parallel to, and why?
4. Prove that a line parallel to two intersecting planes is parallel to their line of intersection.
5. Explain in three sentences why a line and a plane can never share exactly two points.